

GAMBELL, AK, USA

WMO: 702040

Lat: 63.767N

Lon: 171.733W

Elev: 28

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26703

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-10.9	-27.2	1.2	-3.9	-24.1	1.4	-6.1	50.0	1.1	47.3	0.1	17.0	40	1.122

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	54.5	51.2	53.4	50.7	52.0	49.6	52.2	53.7	50.8	52.3	49.6	51.3	12.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.1	55.7	52.9	49.3	52.1	51.6	48.0	49.5	50.6	21.4	53.4	20.6	52.4	19.8	51.1	57.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 43.9	40.1	36.1	DB	-16.5	59.6	6.0	3.1	-20.9	61.9	-24.4	63.7	-27.8	65.5	-32.2	67.8	(4)
(5)			WB	-16.8	52.7	6.0	3.2	-21.1	55.0	-24.6	56.9	-27.9	58.7	-32.2	61.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	7.9	11.4	11.7	18.8	31.1	40.1	46.6	47.3	42.7	34.3	26.1	13.5		
	DBStd	16.47	13.00	13.49	12.15	8.49	4.64	4.34	3.26	3.27	4.38	5.29	7.28	12.23		
	HDD50	8155	1306	1082	1187	936	584	297	113	95	221	488	716	1130		
	HDD65	13609	1771	1502	1652	1386	1049	747	569	549	670	953	1166	1595		
	CDD50	23	0	0	0	0	0	1	10	11	1	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	18.5	23.2	23.1	19.1	18.7	14.5	13.1	13.2	14.7	16.6	19.5	22.5	24.2		
	PrecAvg	22.9	1.7	1.5	1.1	1.0	1.2	1.5	2.4	3.2	2.8	2.4	2.3	1.8		
	PrecMax	34.9	8.1	6.3	3.1	4.0	2.8	3.3	4.2	5.6	4.8	6.0	4.4	4.1		
	PrecMin	13.7	0.0	0.2	0.0	0.2	0.4	0.3	0.7	0.6	1.1	0.5	0.4	0.0		
	PrecStd	4.8	1.6	1.3	0.9	0.7	0.7	0.6	0.8	1.0	1.0	1.2	1.3	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	36.4	34.2	36.1	43.0	54.9	59.0	56.6	52.5	48.9	39.7	35.9		
		MCWB	41.2	34.1	32.6	34.2	38.9	48.1	52.9	52.5	49.5	46.8	37.9	33.8		
	2%	DB	35.9	34.3	32.7	34.1	38.9	51.2	54.9	54.4	51.7	46.4	37.4	33.9		
		MCWB	33.9	32.6	31.3	32.6	36.4	47.1	51.5	51.8	48.9	44.7	35.7	32.6		
	5%	DB	30.4	32.4	31.0	32.3	37.0	48.0	53.5	53.5	49.8	44.8	36.4	31.8		
		MCWB	29.3	31.2	29.7	31.0	34.8	45.3	50.8	51.4	47.5	42.5	34.9	31.0		
	10%	DB	27.3	30.4	29.2	30.1	36.2	45.9	51.8	52.1	48.3	41.1	35.6	29.7		
		MCWB	26.1	29.5	28.0	29.2	34.2	43.4	49.5	50.0	46.0	38.6	34.2	28.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.9	34.8	33.3	34.5	39.5	50.1	54.1	53.9	50.5	47.3	38.4	33.8		
		MCDB	46.4	35.6	33.8	35.1	43.1	53.4	57.1	55.2	51.6	48.3	39.6	34.5		
	2%	WB	33.3	32.6	31.5	32.4	37.1	47.4	52.3	52.5	49.3	45.5	36.6	32.4		
		MCDB	35.0	33.6	32.5	33.4	38.7	50.1	54.2	53.6	50.5	46.7	37.4	33.2		
	5%	WB	29.7	31.5	30.2	31.1	35.6	45.8	50.7	51.2	47.9	42.4	35.1	30.9		
		MCDB	30.5	32.5	31.0	32.2	36.8	47.8	52.6	52.4	49.2	44.7	36.1	31.6		
	10%	WB	25.9	29.7	28.2	29.2	34.3	44.0	49.3	50.0	46.4	39.2	33.5	28.6		
		MCDB	27.3	30.5	29.1	29.9	35.5	46.0	51.3	51.5	47.6	41.5	35.0	29.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.3	8.7	7.1	5.2	6.1	5.4	4.8	4.2	3.9	4.7	6.1		
		MCDBR	13.0	9.2	7.4	7.1	7.1	9.7	9.1	6.4	4.5	3.8	4.3	6.8		
		MCWBR	11.7	9.0	7.0	6.6	5.5	6.6	5.8	4.7	3.7	3.5	4.8	6.9		
	5% WB	MCDBR	13.0	9.1	7.3	6.8	6.5	8.6	7.6	5.4	3.9	3.6	4.4	6.6		
		MCWBR	11.9	8.9	7.0	6.4	5.5	6.1	5.3	4.4	3.6	3.4	5.1	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.185	0.268	0.282	0.348	0.384	0.368	0.379	0.390	0.337	0.304	0.403	0.385		
	taud		1.961	2.075	2.163	2.026	2.149	2.415	2.429	2.342	2.452	2.288	2.675	2.696		
	Ebn at Noon		135	203	256	258	260	270	263	246	239	191	99	60		
	Edh at Noon		9	21	29	42	41	32	31	31	23	17	9	6		
(44) All-Sky Solar Radiation	RadAvg		56	270	801	1307	1417	1674	1350	993	645	266	61	20		
	RadStd		7	41	80	146	152	122	121	106	56	33	7	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page